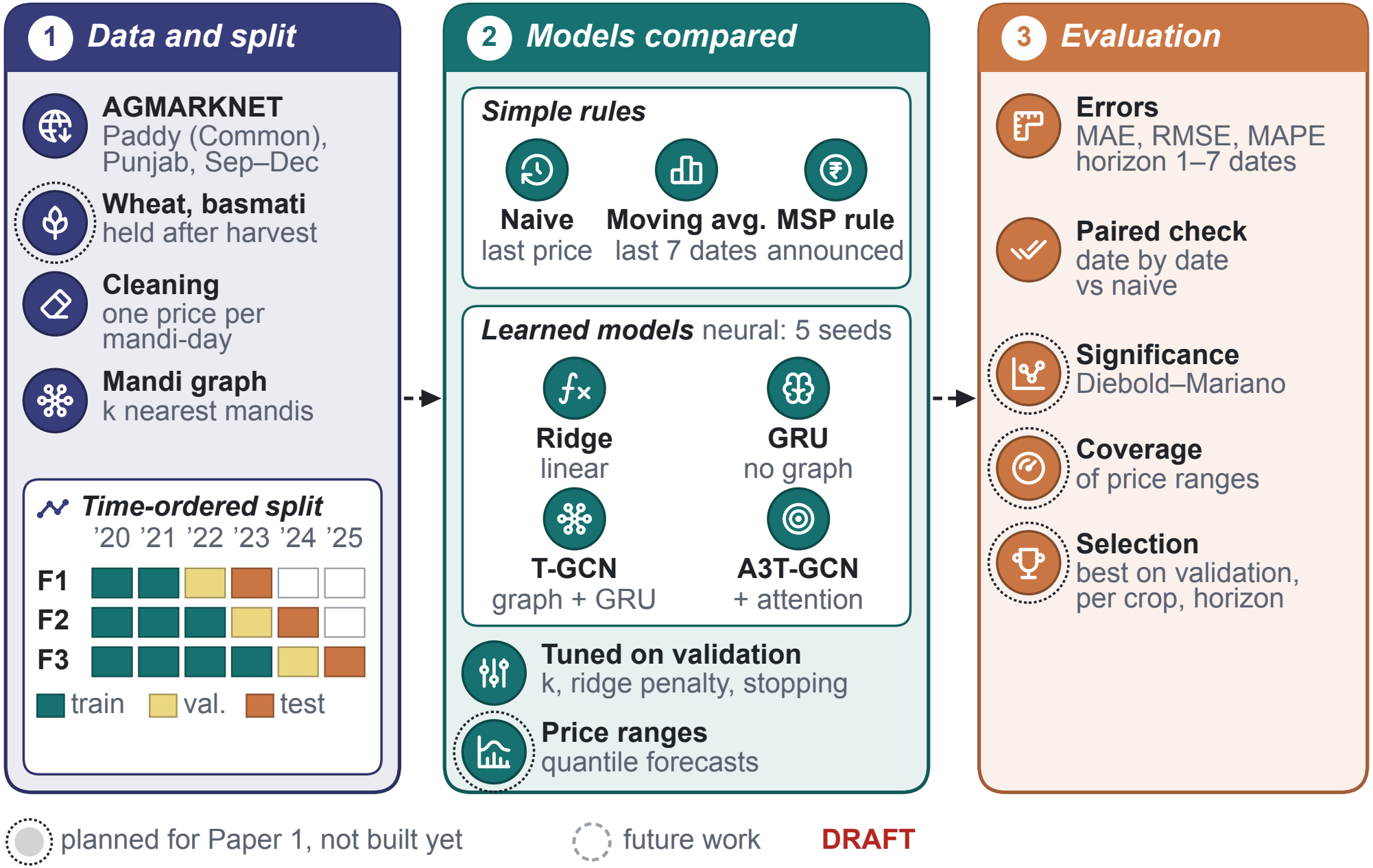

1 Data and split
AGMARKNET
Paddy (Common), Punjab, Sep–Dec
Wheat, basmati
held after harvest
Cleaning
one price per mandi-day
Mandi graph
k nearest mandis
Time-ordered split
'20 '21 '22 '23 '24 '25
F1
F2
F3
train
val.
test
2 Models compared
Simple rules
Naive
last price
Moving avg.
last 7 dates
MSP rule
announced
Learned models neural: 5 seeds
Ridge
linear
GRU
no graph
T-GCN
graph + GRU
A3T-GCN
+ attention
Tuned on validation
k, ridge penalty, stopping
Price ranges
quantile forecasts
3 Evaluation
Errors
MAE, RMSE, MAPE
horizon 1–7 dates
Paired check
date by date
vs naive
Significance
Diebold–Mariano
Coverage
of price ranges
Selection
best on validation,
per crop, horizon
planned for Paper 1, not built yet
future work
DRAFT